

Wazuh SOC Home Lab

Windows Endpoint Monitoring, File Integrity Monitoring, Threat Intelligence,
Firewall Logging, and Network IDS

Hands-On SOC Project Documentation

Overview

This project demonstrates a small Security Operations Center (SOC) environment built in Oracle VirtualBox. Wazuh is used as the central SIEM/XDR platform, with a Windows endpoint providing Sysmon and FIM telemetry, VirusTotal providing reputation context, pfSense providing firewall logs, and Suricata providing network IDS alerts. Kali Linux is used as a controlled traffic-generation system for validation.

The project focuses on building a complete monitoring pipeline rather than a single tool: endpoint activity -> network telemetry -> centralized analysis in Wazuh.

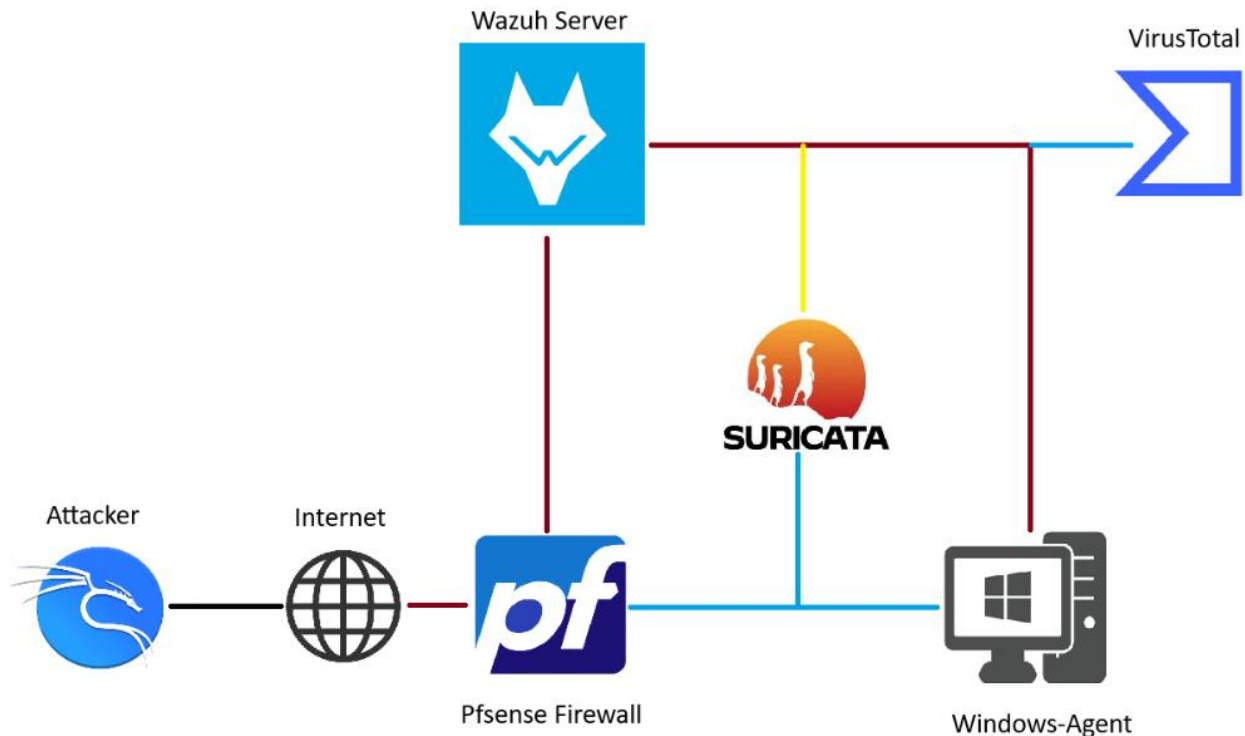
Lab Architecture and Networking

The lab uses two VirtualBox network types: NAT for Internet access and Host-Only for stable communication between the lab systems.

System	Purpose
Windows host / VirtualBox Host-Only adapter	Host access to the isolated lab network
pfSense LAN	Firewall management and syslog source
Wazuh Ubuntu Server	Wazuh Manager, Indexer, Dashboard, syslog receiver
WindowsAgent	Monitored Windows 11 endpoint
Kali Linux	Traffic generation and validation
VirusTotal	Integrated with Api key

VirtualBox NAT addresses such as 10.0.x.x are not used for direct VM-to-VM communication because each VM receives its own isolated NAT environment. All security telemetry and lab-to-lab traffic use the Host-Only addresses.

pfSense is integrated as a monitored firewall/log source. During this build, endpoint VMs retained separate VirtualBox NAT adapters for Internet access while pfSense, Wazuh, WindowsAgent, and Kali communicated over the Host-Only network.



(Lab architecture)

1. Setting Up Wazuh

Wazuh was deployed as the central SIEM/XDR platform for the lab. The Ubuntu Server VM runs the Wazuh Manager, Indexer, Dashboard, and Filebeat, while Windows 11 pro endpoint runs the Wazuh Agent. The goal of this stage was to establish centralized log collection before adding endpoint and network telemetry.

1.1 Wazuh Server Installation

The all-in-one Wazuh deployment was installed on Ubuntu Server using the Wazuh 4.14 installation script.

```
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh
sudo bash ./wazuh-install.sh -a
```

The dashboard was accessed through the Host-Only address:

```
https://192.168.x.x
```

Core services were verified with:

```
sudo systemctl status wazuh-manager
sudo systemctl status wazuh-indexer
sudo systemctl status wazuh-dashboard
sudo systemctl status filebeat
```

```
wazuh-manager      : active
wazuh-indexer      : active
wazuh-dashboard    : active
filebeat           : active
root@wazuh:/home/wazuh# _
```

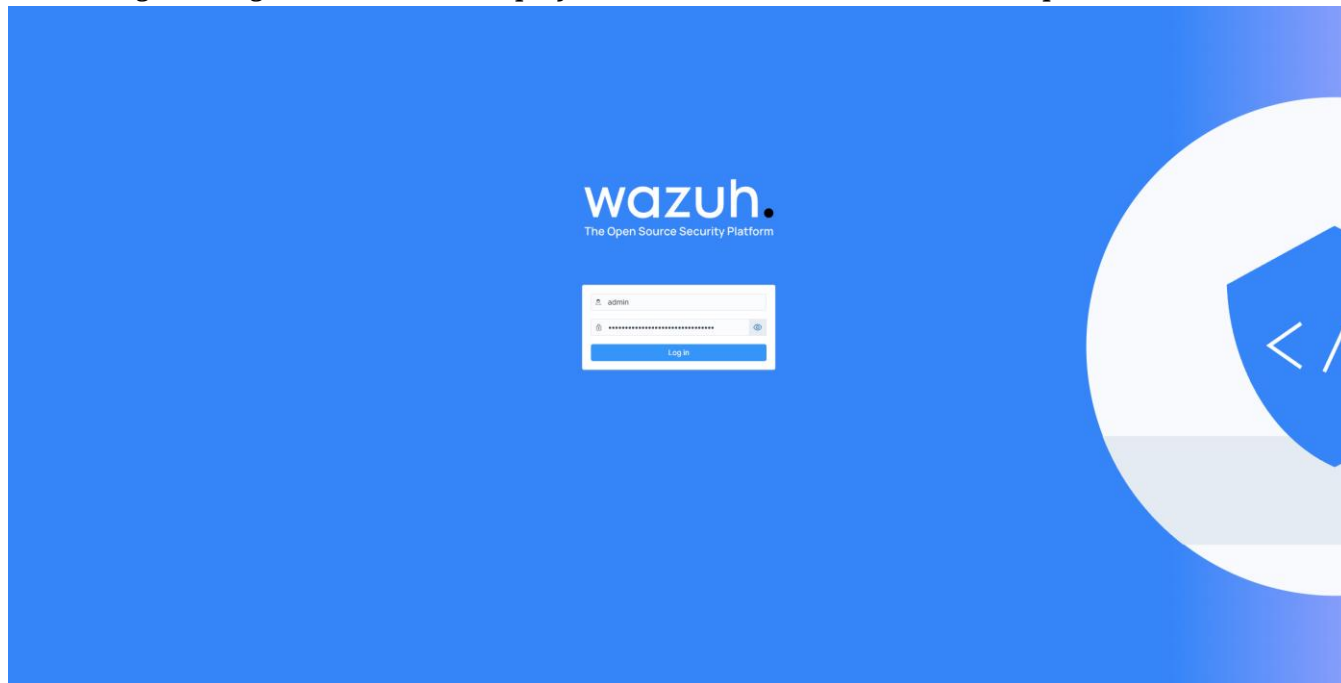
(Statuses of Wazuh services)

1.2 Accessing the Wazuh Dashboard After installation:

Check your Ubuntu VM's IP address: `ifconfig`

On your Ubuntu server, open a browser and go to `https://<ubuntu-vm-ip>:443`

1. Accept any browser security warning due to the self-signed certificate.
2. Log in using the credentials displayed at the end of the installation script



(Wazuh login page)

1.3 Windows Agent Enrollment

The Wazuh Agent was installed and connected via deploy agent tab in wazuh dashboard

< Deploy new agent

1 Select the package to download and install on your system:

LINUX

☐ RPM amd64 ☐ RPM arch64 ☐ DEB amd64 ☐ DEB arch64

WINDOWS

☒ MSI 32/64 bits

macOS

☐ Intel ☐ Apple silicon

For additional systems and architectures, please check our [documentation](#).

2 Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address

Server address

☒ Remember server address

3 Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name

Agent name

The agent name must be unique. It can't be changed once the agent has been enrolled.

Select one or more existing groups

Default

4 Run the following commands to download and install the agent:

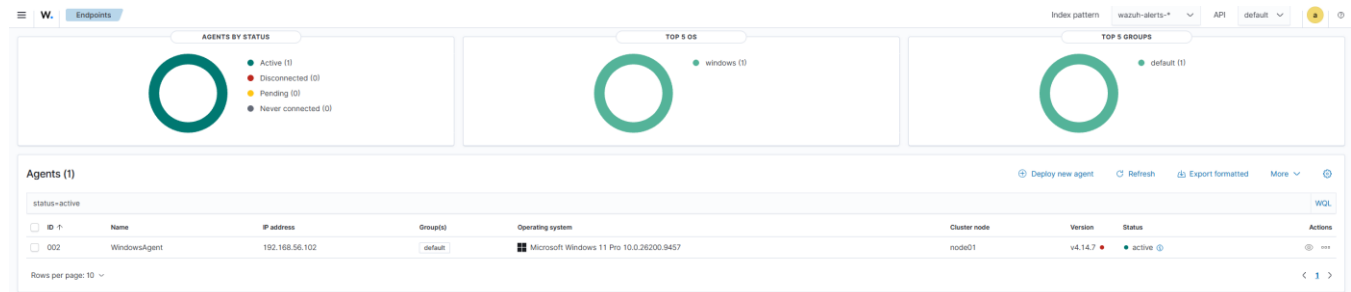
Please select the server address.

(Agent enrollment)

Windows agent was installed and connected with provided command:

```
Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.8-1.msi -OutFile $env:tmp\wazuh-agent; msixexec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='windows-vm-ip' WAZUH_AGENT_NAME='WindowsAgent'
```

After installation we can find connected server in endpoints tab



(endpoints page)

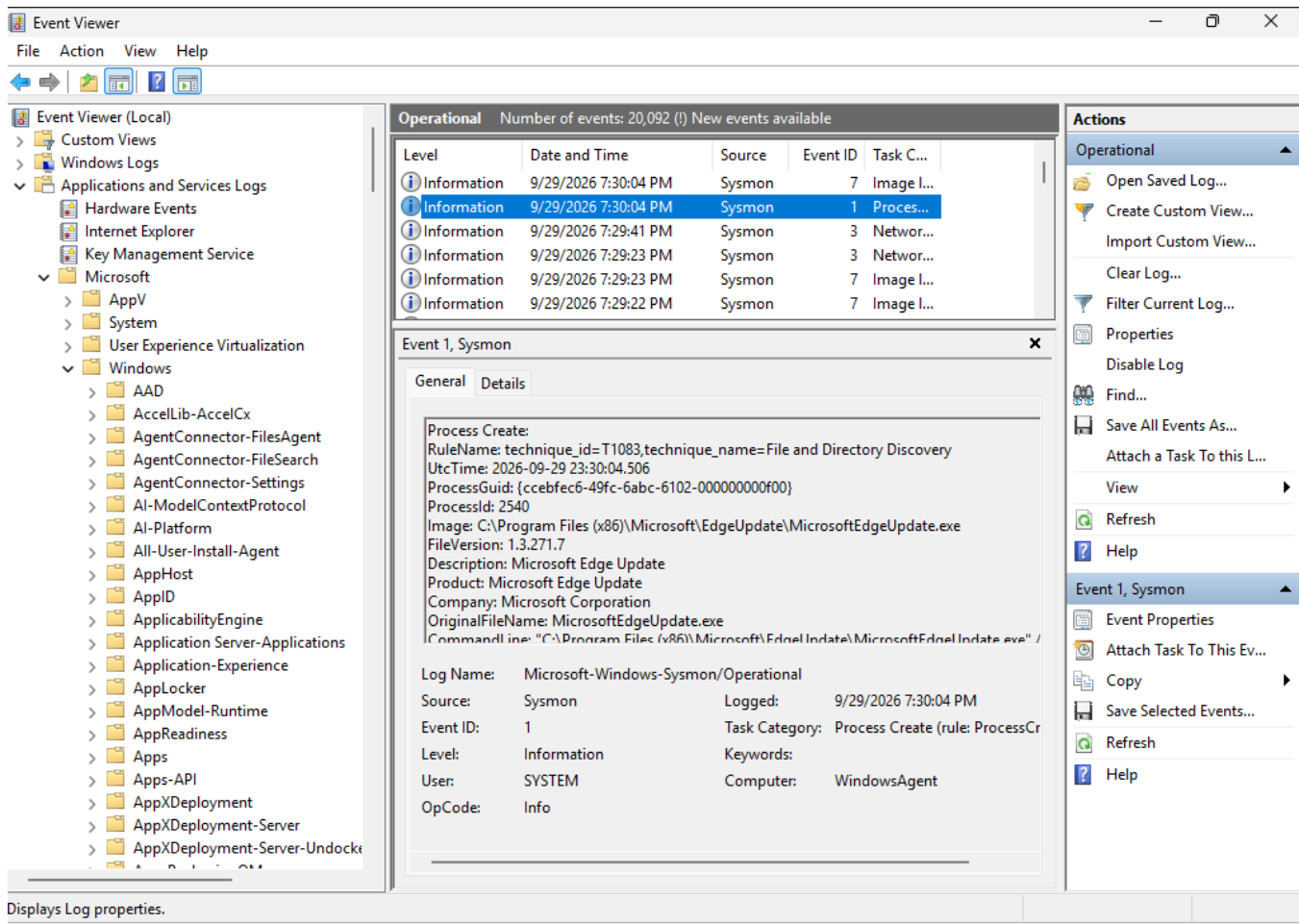
2. Setting Up Sysmon

Installing Sysmon

1. Download from Microsoft Sysinternals.
2. Extract the ZIP.
3. You need to download a configuration file sysmonconfig.xml .
4. Click RAW then Ctrl+S and save the file in the folder where you extracted your zip.
5. In Powershell (run as Administrator) go to the folder where you extracted your Sysmon file. After typing: `./Sysmon64.exe` and tap Enter.
6. In Windows Search Bar type “Services” and find Sysmon. This way we can verify that the installation went correctly.
7. We need to configure Sysmon in Powershell type: `./Sysmon64.exe -i` and Accept agreements.

2.1 Viewing Sysmon Events For double checking Navigate in Event Viewer

Applications and Services Logs -> Microsoft -> Windows -> Sysmon -> Operational



(Sysmon in EventViewer)

2.2 Forward Sysmon Events to Wazuh

The Windows Wazuh Agent was configured to collect the Sysmon Operational event channel.

Default location of the configuration file: C:\Program Files (x86)\ossec-agent\ossec.conf

```
<localfile>
```

```
<location>Microsoft-Windows-Sysmon/Operational</location>
```

```
<log_format>eventchannel</log_format> </localfile>
```

After restarting the Wazuh Agent, Sysmon events became available for investigation in Wazuh.

The Sysmon Operational channel was verified with:

```
Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational" -MaxEvents 5
```

```
PS C:\WINDOWS\system32> Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational" -MaxEvents 5

ProviderName: Microsoft-Windows-Sysmon

TimeCreated      Id LevelDisplayName Message
-----
9/29/2026 7:15:38 PM 15 Information File stream created:...
9/29/2026 7:15:38 PM 7 Information Image loaded:...
9/29/2026 7:15:38 PM 7 Information Image loaded:...
9/29/2026 7:15:38 PM 7 Information Image loaded:...
9/29/2026 7:15:38 PM 17 Information Pipe Created:...
```

(Verified Sysmon Operational Channel)

Verified alert forwarding with eventID = 1 process creation. (launched notepad.exe)

data.win.eventdata.details	Binary Data
data.win.eventdata.eventType	SetValue
data.win.eventdata.image	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2607.14.0_x64__8wekyb3d8bbwe\Notepad\Notepad.exe
data.win.eventdata.processGuid	{ccebfe6-4b6d-6abc-7502-00000000f00}
data.win.eventdata.processId	4976
data.win.eventdata.targetObject	HKLM\System\CurrentControlSet\Services\lsam\State\UserSettings\LS-1-5-21-1833662285-2196112776-19543658-1000\Microsoft.WindowsNotepad_8wekyb3d8bbwe
data.win.eventdata.user	WINDOWSAGENT\windows
data.win.eventdata.utcTime	2026-09-29 23:36:15.547
data.win.system.channel	Microsoft-Windows-Sysmon/Operational

(Sysmon logs)

3. Setting Up File Integrity Monitoring (FIM)

Wazuh File Integrity Monitoring was configured on the Windows endpoint to detect file creation, modification, and deletion in selected directories. This provides host-level visibility into changes that may indicate malware activity, persistence, tampering, or unauthorized user actions.

3.1 Configure Monitored Directories

The following directories were added inside the <syscheck> block of the Windows agent configuration at: C:\Program Files (x86)\ossec-agent\ossec.conf

```
<directories realtime="yes">C:\Users\Windows\Desktop\test</directories>
```

The agent was restarted after the configuration change.

3.2 Validate FIM

Test files were created, edited, and deleted inside the monitored directory. The resulting file integrity events were reviewed in the Wazuh dashboard.

9 hits						
Sep 29, 2026 @ 19:53:44.441 - Sep 29, 2026 @ 19:53:44.441						
Export Formatted	Reset view	812 available fields	Columns	Density	1 fields sorted	Full screen
timestamp	agent.name	syscheck.path	syscheck.event	rule.description	rule.level	rule.id
Sep 29, 2026 @ 19:52:35.9...	WindowsAgent	c:\users\windows\desktop\test\new text document.txt	deleted	File deleted.	7	553
Sep 29, 2026 @ 19:52:34.0...	WindowsAgent	c:\users\windows\desktop\test\new text document.txt	modified	Integrity checksum changed.	7	550
Sep 29, 2026 @ 19:52:29.6...	WindowsAgent	c:\users\windows\desktop\test\new text document.txt	modified	Integrity checksum changed.	7	550
Sep 29, 2026 @ 19:52:27.0...	WindowsAgent	c:\users\windows\desktop\test\new text document.txt	modified	Integrity checksum changed.	7	550
Sep 29, 2026 @ 19:52:23.8...	WindowsAgent	c:\users\windows\desktop\test\new text document.txt	added	File added to the system.	5	554

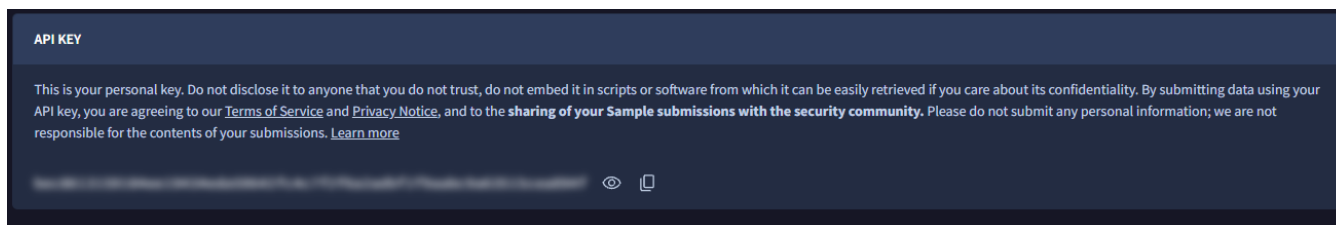
(Events in FIM)

4. Setting Up VirusTotal Integration

VirusTotal was integrated with Wazuh to add threat-intelligence context to file integrity events. When a monitored file changes, Wazuh can query VirusTotal using the file hash and return reputation information to the analyst.

4.1 Obtain API key

Obtain a VirusTotal API Key. You need to have an account in VirusTotal, then go to Profile and copy your API KEY.



(Virus total API page)

4.2 Configure the Wazuh Manager

A VirusTotal API key was added to the Wazuh Manager configuration at: /var/ossec/etc/ossec.conf

```
<integration>
  <name>virustotal</name>
  <api_key>[REDACTED]</api_key>
  <group>syscheck</group>
  <alert_format>json</alert_format>
</integration>

</ossec_config>
```

(VirusTotal configuration on Wazuh manager)

The integration was placed on the Wazuh Manager because integrations are executed on server-side.

4.3 Configure wazuh agent

Open configuration file at: C:\Program Files (x86)\ossec-agent\ossec.conf and add

```
<directories realtime="yes">C:\Users\Windows\Desktop\virustotal</directories>
```

This ensures that any file added to Desktop\virustotal will trigger a VirusTotal lookup. Restart the agent to apply for changes.

4.4 Validate the Integration

A test file was created in the monitored Virustotal folder. Wazuh generated the FIM event and enriched the event with VirusTotal lookup information.

Check configuration Create a file in the C:\Users\Windows\Desktop\virustotal directory to check if our connection with Wazuh works. Then go to Wazuh Dashboard - Discover and find our log. Our event also will appear in the File Integrity section. In Discover, we can see a link to VirusTotal where we can become more information about our file.

data.virustotal.permlink	https://www.virustotal.com/gui/file/468ffe129c395abf6b21a09efdf261910a95fb98aa982ead73caa7b2b684577e/detection/f-468ffe129c395abf6b21a09efdf261910a95fb98aa982ead73caa7b2b684577e-1790782157
data.virustotal.positives	0
data.virustotal.scan_date	2026-09-29 17:15:57
data.virustotal.sha1	76cd26b59923157e09d2bc927ba8fb059f3155dc
data.virustotal.source.alert_id	1790728656.165083
data.virustotal.source.file	C:\users\windows\desktop\virustotal\notepad-test.exe

(VirusTotal link in logs)

5. Setting Up pfSense Firewall Logging

A firewall serves as a foundational security tool that monitors and manages data entering or leaving a network. Much like a checkpoint guard controlling entry to a facility, it permits or denies traffic

according to established security guidelines. Within today's digital infrastructures, firewalls are essential for:

- Restricting unapproved connections
- Filtering out dangerous cyber threats
- Dividing networks into isolated zones

Firewall technologies range from basic packet filtering and stateful inspection to proxy gateways and next-generation firewalls (NGFWs), with each type offering deeper traffic visibility. This tutorial walks you through setting up a practical SOC sandbox utilizing two primary tools: pfSense for open-source routing and network defense, and Wazuh as the centralized SIEM for threat detection and log aggregation. Connecting pfSense telemetry to Wazuh allows you to replicate corporate SOC workflows and achieve comprehensive visibility over network events

5.1 Requirements

Before starting, ensure you have the following:

1. Virtual Machine Software → VMware Workstation Pro (used in this guide, but VirtualBox also works).
2. pfSense ISO → Download the latest version from the official pfSense website.
3. Wazuh Manager → Running and accessible in your lab environment.

5.2 Creating Virtual Machine for pfsense

Open VMware Navigate to File → New Virtual Machine.

2. Select the Installation Media Choose Installer disc image file (ISO) and browse for the downloaded pfSense ISO.
3. Name Your VM Use a descriptive name such as pfSense and pick a location to save your VM.
4. Allocate Disk Space • 20 GB minimum (store as a single file recommended)

5.3 Customize Hardware

- RAM: 1 GB minimum (2 GB recommended).
- CPU: 1 vCPU (2 cores recommended).
- Network Adapters: – Adapter 1 (WAN) → NAT (simulates internet access).
- – Adapter 2 (LAN) → Host-only (isolated internal lab network).

5.4 Assigning interfaces

- WAN (em0) → Configured via DHCP (IP in 192.x.x.x/24 range).
- LAN (em1) → Static IP 192.168.1.1/24.

```
pfSense [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Starting package System Patches...done.
pfSense 2.9.0-RELEASE amd64 20260925-1514
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)
VirtualBox Virtual Machine - Netgate Device ID: 35d9731b641fbadd76db
*** Welcome to pfSense 2.9.0-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 
v6/DHCP6: 
LAN (lan) -> em1 -> v4: 

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: 
```

(pfsense interfaces)

5.5 Accessing the pfSense Web Interface

1. Open a browser from a VM (e.g., Kali Linux) connected to the LAN network.
2. Go to pfsense login page (pfsense host ip)
3. Log in with default credentials:
 - Username: admin
 - Password: pfsense

Important: Change the default credentials immediately after login



Login to pfSense

A screenshot of the pfSense login page. The background is a solid dark blue. In the center, the text 'SIGN IN' is displayed in white. Below it are two input fields: the first is labeled 'Username' and the second is labeled 'Password', both in a light blue font. A green 'SIGN IN' button is positioned below the password field.

(Pfsense Login page)

Status / Dashboard

System Information

Name	pfSense.home.arpa
User	admin@ (Local Database)
System	VirtualBox Virtual Machine Netgate Device ID: 35d9731b641fbadd76db
BIOS	Vendor: innotek GmbH Version: VirtualBox Release Date: Fri Dec 1 2006 Boot Method: BIOS
Version	2.9.0-RELEASE (amd64) built on Fri Sep 25 15:14:00 UTC 2026 FreeBSD 16.0-CURRENT The system is on the latest version. Version information updated at Wed Sep 30 1:27:08 UTC 2026
CPU Type	AMD Ryzen 5 9600X 6-Core Processor AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive
Kernel PTI	Disabled
MDS Mitigation	Inactive
Uptime	00 Hour 01 Minute 49 Seconds
Current date/time	Wed Sep 30 1:28:03 UTC 2026
DNS server(s)	
Last config change	Wed Sep 30 1:23:45 UTC 2026
State table size	0% (90/403000) Show states
MBUF Usage	0% (3556/1000000)
Load average	1.15, 0.71, 0.30
CPU usage	5%
Memory usage	10% of 4036 MiB
SWAP usage	0% of 1024 MiB

Netgate Services And Support

Contract Type

Community Support
Community Support Only

NETGATE & PFSense COMMUNITY SUPPORT RESOURCES

If you purchased your pfSense gateway firewall appliance from Netgate and elected **Community Support** at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the [NETGATE RESOURCE LIBRARY](#).

You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space.

- Upgrade Your Support
- Community Support Resources
- Netgate Global Support FAQ
- Official pfSense Training by Netgate
- Netgate Professional Services
- Visit Netgate.com

If you decide to purchase a Netgate Global TAC Support subscription, you **MUST** have your **Netgate Device ID (NDI)** from your firewall in order to validate support for this unit. Write down your NDI and store it in a safe place. You can purchase a TAC support plan [here](#).

Interfaces

WAN	1000baseT <full-duplex>
LAN	1000baseT <full-duplex>

(pfsense Dashboard)

5.6 Why Combine pfSense with Wazuh?

Combining pfSense and Wazuh bridges the gap between network security and system monitoring. Together, they create a complete, open-source Security Operations Center (SOC) solution. Here is why this integration is important for the project:

- **All-in-One Dashboard:** Instead of checking pfSense and servers separately, all logs are collected in one central Wazuh interface.
- **Smart Alerting:** pfSense generates thousands of raw logs. Wazuh automatically analyzes them and highlights only the real threats, like DDoS attacks or port scans.

- **Faster Incident Response:** If Wazuh detects suspicious behavior inside the network, it can automatically tell pfSense to block the attacker's IP address.
- **Attack Correlation:** Wazuh can connect the dots between an external network connection logged by pfSense and a failed login attempt logged by an internal server.
- **Long-Term Storage:** pfSense has limited memory for logs. Wazuh safely archives all firewall history for future security analysis and compliance.

5.7 Configuring pfSense Logs in Wazuh

Enable Remote Logging in pfSense

- Go to Status → System Logs → Settings.
- Check Enable Remote Logging.
- Add your Wazuh server's IP (e.g., 192.168.x.x) under Remote Syslog Servers.
- Set Port to 514 (default syslog).
- Select log categories to forward: System, Firewall, VPN, DHCP, DNS.

Remote Logging Options

Enable Remote Logging

☒ Send log messages to remote syslog server

Source Address

LAN

This option will allow the logging daemon to bind to a single IP address, rather than all IP addresses. If a single IP is picked, remote syslog servers must all be of that IP type. To mix IPv4 and IPv6 remote syslog servers, bind to all interfaces.

NOTE: If an IP address cannot be located on the chosen interface, the daemon will bind to all addresses.

IP Protocol

IPv4

This option is only used when a non-default address is chosen as the source above. This option only expresses a preference; If an IP address of the selected type is not found on the chosen interface, the other type will be tried.

Remote log servers

514

IP[:port]

IP[:port]

Remote Syslog Contents

☒ Everything

☐ System Events

☐ Firewall Events

☐ DNS Events (Resolver/unbound, Forwarder/dnsmasq, filterdns)

☐ DHCP Events (DHCP Daemon, DHCP Relay, DHCP Client)

☐ PPP Events (PPPoE WAN Client, L2TP WAN Client, PPTP WAN Client)

☐ General Authentication Events

☐ Captive Portal Events

☐ VPN Events (IPsec, OpenVPN, L2TP, PPPoE Server)

☐ Gateway Monitor Events

☐ Routing Daemon Events (RADVD, UPnP, RIP, OSPF, BGP)

☐ Network Time Protocol Events (NTP Daemon, NTP Client)

☐ Wireless Events (hostapd)

Syslog sends UDP datagrams to port 514 on the specified remote syslog server, unless another port is specified. Be sure to set syslogd on the remote server to accept syslog messages from pfSense.

(pfSense System Logs Settings)

5.8 Configure Wazuh Syslog

Input On the Wazuh Manager, edit /var/ossec/etc/ossec.conf:

Wazuh SOC Home Lab

```
<!--pfSense syslog input -->
<remote>
  <connection>syslog</connection>
  <port>514</port>
  <protocol>udp</protocol>
  <allowed-ips>[redacted]</allowed-ips>
  <local_ip>[redacted]</local_ip>
</remote>
```

(syslog configuration)

Restart Wazuh after changes

5.9 Creating a Custom Decoder and Rules for pfSense Logs

Since pfSense logs have a unique format, Wazuh may not parse them correctly out of the box.

You can add a new decoder in the Wazuh Dashboard in Server management - Decoders .

Decoders (1,578)

From here you can manage your decoders.

Search

Manage decoders files Add new decoders file Refresh Export formatted

Name	Program name	Order	File	Path
wazuh			0005-wazuh_decoders.xml	ruleset/decoders
agent-buffer		level	0005-wazuh_decoders.xml	ruleset/decoders
agent-upgrade		agent_id, agent_name, status	0005-wazuh_decoders.xml	ruleset/decoders
agent-upgrade		error	0005-wazuh_decoders.xml	ruleset/decoders
agent-upgrade		agent_cur_version	0005-wazuh_decoders.xml	ruleset/decoders
agent-upgrade		agent_new_version	0005-wazuh_decoders.xml	ruleset/decoders
agent-restart		module	0005-wazuh_decoders.xml	ruleset/decoders
fm-state			0005-wazuh_decoders.xml	ruleset/decoders
json-mqgraph			0006-json_decoders.xml	ruleset/decoders
json			0006-json_decoders.xml	ruleset/decoders

Rows per page: 10

(decoders page)

A decoder extracts key fields (source IP, destination IP, protocol). Example (pfsense-custom-decoder.xml):

< Pfsense.xml

```
1 <!-- Modify it at your will. -->
2 <decoder name="pfsense-custom">
3   <prematch>filterlog</prematch>
4 </decoder>
5
6 <decoder name="pfsense-fields">
7   <parent>pfsense-custom</parent>
8
9   <regex>(\w+)\d+:\S*,\S*,\S*,(\S+),\S*,\S*,(\S+),\S*,\S*,\S*,\S*,\S*,\S*,\S*,(\S+),\S*,\S*,(\S+),(\S+),(\d+);(\d+),\S*</regex>
10
11   <order>logsource,id,section,protocol,srcip,dstip,srcport,dstport</order>
12
13 </decoder>
```

(custom decoder)

5.10 Custom Rules

A rule assigns severity levels and alerts based on matched events. You can add a new rule in the Wazuh Dashboard in Server management – Rules

Rules (4,518) Manage rules files Add new rules file Refresh Export formatted

From here you can manage your rules.

ID ↑	Description	Groups	Level	File	Path
1	Generic template for all syslog rules.	syslog	0	0010-rules_config.xml	ruleset/rules
2	Generic template for all firewall rules.	firewall	0	0010-rules_config.xml	ruleset/rules
3	Generic template for all ids rules.	ids	0	0010-rules_config.xml	ruleset/rules
4	Generic template for all web rules.	web-log	0	0010-rules_config.xml	ruleset/rules
5	Generic template for all web proxy rules.	squid	0	0010-rules_config.xml	ruleset/rules
6	Generic template for all windows rules.	windows	0	0010-rules_config.xml	ruleset/rules
7	Generic template for all wazuh rules.	ossec	0	0010-rules_config.xml	ruleset/rules
200	Grouping of wazuh rules.	wazuh	0	0018-wazuh_rules.xml	ruleset/rules
201	Agent event queue rule	agent_flooding, wazuh	0	0018-wazuh_rules.xml	ruleset/rules
202	Agent event queue is level full.	agent_flooding, wazuh	7	0018-wazuh_rules.xml	ruleset/rules

Rows per page: 10 < 1 2 3 4 5 ... 452 >

(rules page)

Example (pfsense-custom-rules.xml):

< pfsense-custom-decoder.xml

```
1
2 <group name="pfsense,firewall,">
3   <!-- Generic pfSense log -->
4   <rule id="100111" level="3">
5     <decoded_as>pfsense-custom</decoded_as>
6     <description>pfSense log received</description>
7   </rule>
8   <!-- Successful login -->
9   <rule id="100112" level="3">
10    <if_sid>100111</if_sid>
11    <match>Successful login</match>
12    <description>pfSense: Successful login from
13    $(srcip)</description>
14  </rule>
15  <!-- Allowed traffic -->
16  <rule id="100113" level="5">
17    <if_sid>100111</if_sid>
18    <match>pass</match>
19    <description>pfSense: Allowed traffic from $(srcip) to
20    $(dstip)</description>
21  </rule>
22  <!-- Blocked traffic -->
23  <rule id="100114" level="7">
24    <if_sid>100111</if_sid>
25    <match>block</match>
26    <description>pfSense: Blocked traffic from $(srcip) to
27    $(dstip)</description>
28  </rule>
29  <!-- Authentication error -->
30  <rule id="100115" level="10">
31    <if_sid>100111</if_sid>
32    <match>authentication error</match>
33    <description>pfSense: Authentication error for user
34    $(dstuser) from $(srcip)</description>
35  </rule>
36 </group>
```

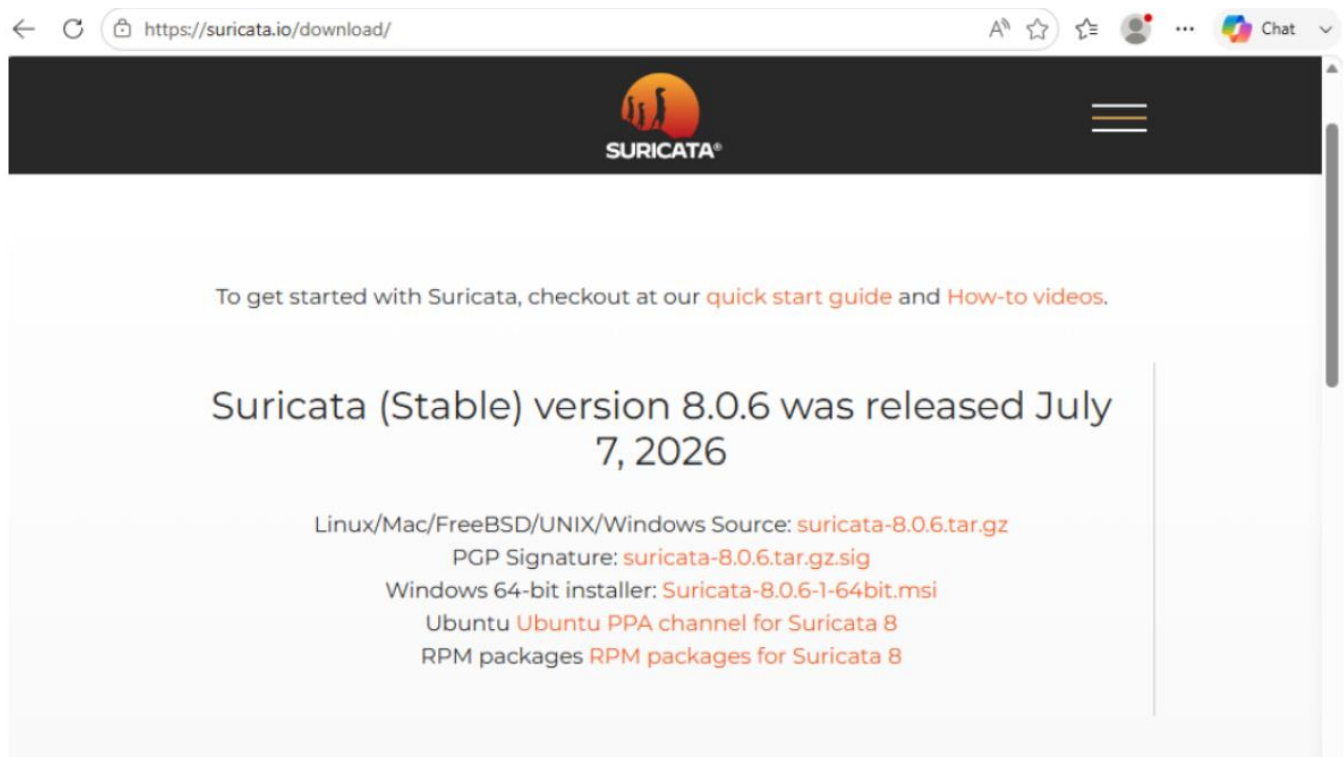
(custom rule)

6. Setting Up Suricata IDS/IPS with a Custom Rule

Suricata was installed on the Windows endpoint as the network detection engine. Suricata supports both IDS and IPS operations; this lab currently uses IDS mode, where suspicious traffic is detected and logged but not blocked. Alerts are written to eve.json and forwarded to Wazuh by the Windows agent.

6.1 Install and Configure Suricata

1. Visit: <https://suricata.io/download/>
2. Download the Windows installer (.msi)
3. Run the installer with default settings Suricata installs to: C:\Program Files\Suricata\ Contents include:
 - suricata.exe – the main executable
 - suricata.yaml – configuration file
 - rules\ – rule files directory
 - log\ – log output folder



(Suricata Download Page)

6.2 Install and Verify Npcap

Suricata requires Npcap to capture network traffic.

1. Download from <https://npcap.com/#download> During installation:
 - Enable WinPcap API-compatible mode
 - Enable startup at boot

← ↻ 🔒 https://npcap.com/#download

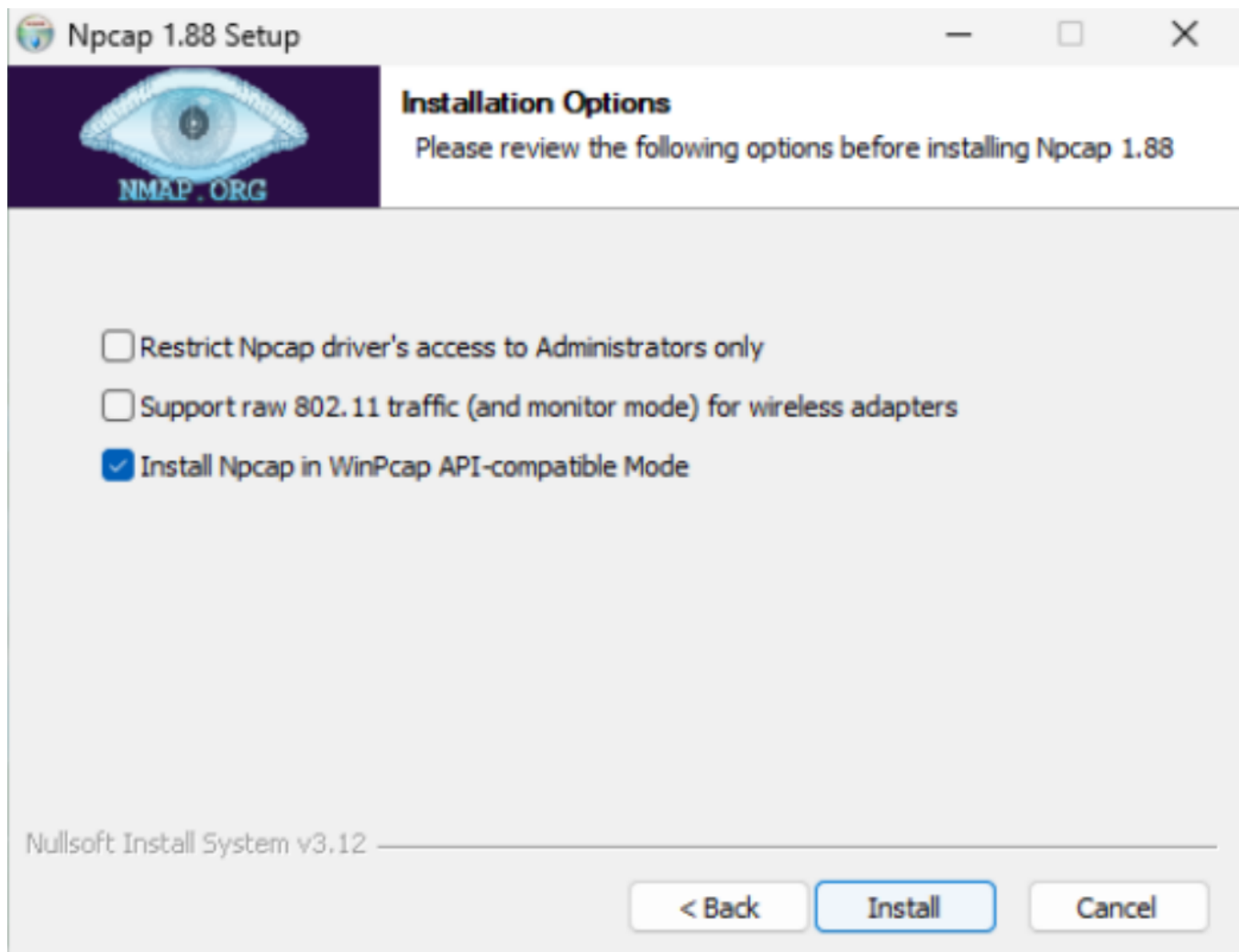
Downloading and Installing Npcap Free Edition

The free version of Npcap may be used (but not externally redistributed) on up to 5 systems ([free license details](#)). It may also be used on unlimited systems where it is only used with [Nmap](#), [Wireshark](#), and/or [Microsoft Defender for Identity](#). Simply run the executable installer. The full source code for each release is available, and developers can build their apps against the SDK. The improvements for each release are documented in the [Npcap Changelog](#).

- [Npcap 1.88 installer](#) for Windows 7/2008R2, 8/2012, 8.1/2012R2, 10/2016, 2019, 11 (x86, x64, and ARM64).
- [Npcap SDK 1.16](#) (ZIP).
- [Npcap 1.88 debug symbols](#) (ZIP).
- [Npcap 1.88 source code](#) (ZIP).

The latest development source is in our [Github source repository](#). Windows XP and earlier are not supported; you can use [WinPcap](#) for these versions.

(Npcap Download Page)



(Enable WinPcap API-compatible mode)

2. Verify that Npcap is running:

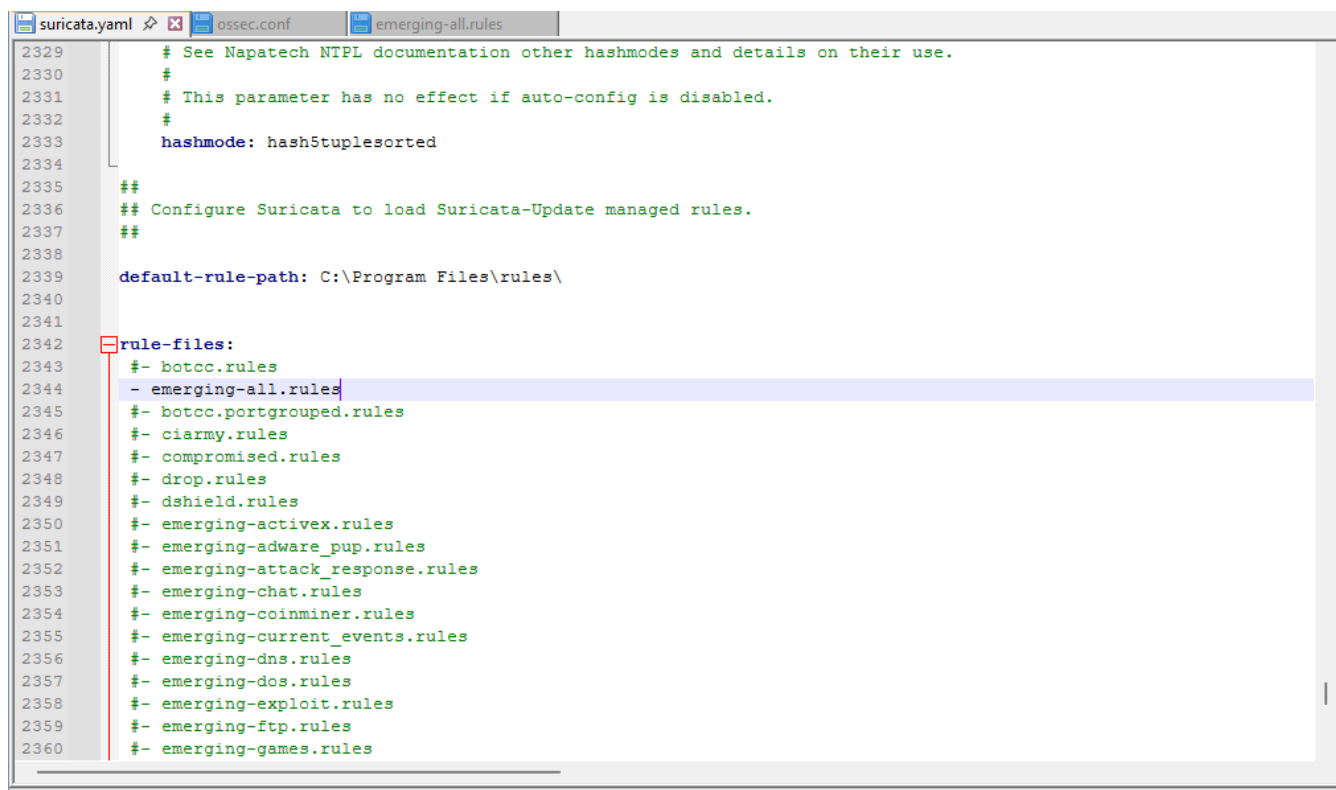
```
PS C:\WINDOWS\system32> get-service -name npcap

Status      Name      DisplayName
-----
Running     npcap     Npcap Packet Driver (NPCAP)
```

(npcap check)

6.3 Configure Suricata

1. Download emerging-all.rules from [https:// rules.emergingthreats.net/open/suricata-7.0.3/](https://rules.emergingthreats.net/open/suricata-7.0.3/) (Rule download)
2. Open Configuration C:\Program Files\Suricata\suricata.yaml Add the emerging-all.rules rule and comment out all other rules. In our home environment, we need only this rule;



```
2329      # See Napatech NTPL documentation other hashmodes and details on their use.
2330      #
2331      # This parameter has no effect if auto-config is disabled.
2332      #
2333      hashmode: hashStuplesorted
2334
2335      ##
2336      ## Configure Suricata to load Suricata-Update managed rules.
2337      ##
2338
2339      default-rule-path: C:\Program Files\rules\
2340
2341
2342      rule-files:
2343      #- botcc.rules
2344      - emerging-all.rules
2345      #- botcc.portgrouped.rules
2346      #- ciarmy.rules
2347      #- compromised.rules
2348      #- drop.rules
2349      #- dshield.rules
2350      #- emerging-activex.rules
2351      #- emerging-adware_pup.rules
2352      #- emerging-attack_response.rules
2353      #- emerging-chat.rules
2354      #- emerging-coinminer.rules
2355      #- emerging-current_events.rules
2356      #- emerging-dns.rules
2357      #- emerging-dos.rules
2358      #- emerging-exploit.rules
2359      #- emerging-ftp.rules
2360      #- emerging-games.rules
```

(suricata.yaml file)

3. Update Wazuh Agent Configuration on Wazuh Agent:

C:\Program Files(x86)\ossec-agent\ossec.conf

```
<localfile>
  <log_format>json</log_format>
  <location>C:\Program Files\Suricata\log\eve.json</location>
</localfile>

</ossec_config>
```

(configuration file)

6.4 Create a Custom ICMP Detection Rule

A local rule was added to create a predictable alert whenever the Windows endpoint receives an ICMP echo request. This provides a simple end-to-end validation of network detection and Wazuh ingestion.

```
alert icmp any any -> any any (msg:"LAB ICMP ping detected"; itype:8; sid:1000001; rev:1;)
```

(custom rule in emerging-all.rules file)

6.5 Run Suricata on the Windows Host-Only Interface

Suricata was started manually against the WindowsAgent Host-Only interface:

```
.\suricata.exe -c .\suricata.yaml -i <windows-agent-ip>
```

The PowerShell window must remain open while Suricata is running manually. The output line "Engine started" confirms that the sensor is active.

```
PS C:\WINDOWS\system32> cd "C:\Program Files\Suricata"
PS C:\Program Files\Suricata> .\suricata.exe -c .\suricata.yaml -i [redacted]
Info: win32-service: Running as service: no
Info: suricata: translated [redacted] to pcap device \Device\NPF_{E70CD53A-4DBE-4302-8542-DC922C59A083}
i: suricata: This is Suricata version 8.0.7 RELEASE running in SYSTEM mode
i: runmodes: thread stack size of 0 too small: setting to 512k
W: threshold-config: Error opening file: "C:\Program Files\Suricata\\threshold.config": No such file or directory
W: detect-flowbits: flowbit 'ET.Winos4.Login' is checked but not set. Checked in 2059975 and 0 other sigs
W: suricata: setrlimit unavailable.
i: threads: Threads created -> RX: 1 W: 2 FM: 1 FR: 1 Engine started.
```

(suricata running manually)

6.6 Generate and Verify the Alert

Traffic was generated from another lab VM:

```
ping <windows-agent-ip>
```

```
root@wazuh:/home/wazuh# ping 192.168.
```

(pinging host)

The alert was first verified locally on eve.json and then in Wazuh Threat Hunting.

	Sep 29, 2026 @ 21:00:37.6...	WindowsAgent	Suricata: Alert - LAB ICMP ping detected
	Sep 29, 2026 @ 21:00:37.6...	WindowsAgent	Suricata: Alert - LAB ICMP ping detected
	Sep 29, 2026 @ 21:00:35.6...	WindowsAgent	Suricata: Alert - LAB ICMP ping detected
	Sep 29, 2026 @ 21:00:35.6...	WindowsAgent	Suricata: Alert - LAB ICMP ping detected
	Sep 29, 2026 @ 20:58:04.9...	WindowsAgent	Suricata: Alert - ET POLICY Spotify P2P Client

(Suricata alert in Wazuh dashboard)

7. End-to-End Validation

The completed lab was validated by checking each stage of the monitoring pipeline.

Validation	Result
Wazuh Manager and dashboard available	Central SIEM services operational
WindowsAgent active in Wazuh	Endpoint telemetry successfully enrolled

Sysmon events visible in Wazuh	Enhanced Windows telemetry successfully ingested
FIM detects file create/modify/delete activity	Host file monitoring operational
VirusTotal enriches FIM events	Threat-intelligence enrichment operational
pfSense sends UDP syslog to Wazuh	Firewall logging centralized
Suricata alerts from eve.json appear in Threat Hunting	Network IDS telemetry operational
Custom ICMP alert generated from cross-VM ping	End-to-end detection pipeline validated

8. Scaling the Lab: This is Just the Beginning

While this initial project establishes a foundation for threat detection, it represents only a fraction of what a matured security home lab can achieve. To transition this environment into an enterprise-grade defensive ecosystem, I plan to implement the following improvements in my future work:

- **Move from IDS to IPS (Active Prevention):** Change Suricata's configuration from passive detection (action: allowed) to active blocking mode to drop malicious scanning traffic automatically at the firewall level.
- **Configure Wazuh Active Response:** Set up automated containment scripts on endpoints to block attacking source IPs via host firewalls or isolate compromised machines immediately upon high-level alert triggers.
- **Integrate Threat Intelligence Feeds:** Connect the SIEM and IDS engines with open-source threat intelligence (such as MISP or AlienVault OTX) to automatically flag known malicious IP addresses and indicators of compromise (IoCs).
- **Deploy Honeypots:** Introduce a low-interaction honeypot within an isolated DMZ zone to safely capture, log, and study live attacker techniques.

This project has been an invaluable milestone as my first practical experience in infrastructure security. It allowed me to deeply understand log analysis, tool integration, and defensive architecture, providing a solid foundation for my journey into Cyber Security.

9. Final Summary

This project marks my very first hands-on cybersecurity lab and gives me the opportunity to apply concepts that I had previously only studied in theory. By configuring and integrating Wazuh SIEM/XDR, Sysmon, File Integrity Monitoring, VirusTotal, pfSense, and Suricata, I was able to build a complete security monitoring environment and better understand how these tools work together in a SOC setting.

One of the most valuable parts of this project was troubleshooting the issues that appeared during the setup. Problems with networking, log collection, service configuration, and integrations helped me understand the environment in much more detail than a perfectly smooth installation would have. Successfully detecting network activity and Suricata alerts in Wazuh confirmed that the lab was functioning as intended.

I genuinely enjoyed working on this lab, especially seeing each component begin to communicate with the others and watching the final alerts appear in the Wazuh dashboard. It was a challenging but rewarding experience, and it gave me more confidence in working with SIEM tools, endpoint monitoring, network security, and threat detection. This lab also gave me a strong foundation that I can continue expanding with additional detection rules, attack simulations, and incident-response exercises.